Practical Python and Azure DevOps

Four-Day Training and Implementation Plan

Day 1 – Python Programming Fundamentals

1. Introduction to Python and Development Setup
2. Variables, Data Types, Lists, and Dictionaries
3. Conditional Statements and Loops
4. Functions, Modules, and File Handling
5. Introduction to Classes and Objects
6. Hands-on Exercise: Processing Operational Data
7. Day 1 Review and Practical Assessment

Deliverable: A working Python script with reusable functions, sample data processing, and output generation.

Day 2 – API Integration and Data Processing

1. Introduction to APIs and SQL-Backed Data
2. HTTP Requests, Endpoints, and JSON Responses
3. Calling Read-Only APIs Using Python
4. Data Transformation, Validation, and CSV/JSON Export
5. Exception Handling, Logging, and Troubleshooting
6. Introduction to Unit Testing with pytest
7. Hands-on Exercise: API Data Retrieval and Processing

Deliverable: A Python-based API data retrieval and processing script with basic error handling and unit tests.

Day 3 – Azure DevOps and Release Management

1. Introduction to Git, Azure Repos, and Version Control
2. Branching, Commits, Pull Requests, and Code Reviews
3. Introduction to Azure Pipelines and YAML
4. Automated Validation and Test Execution
5. Pipeline Artifact Creation and Publication
6. Environment Approvals and Release Controls
7. Report/Model Deployment Workflow and Network-Team Handoff
8. Release Validation, Verification, and Recovery Planning

Deliverable: A demonstrated Azure DevOps pipeline with validation, artifact publication, approval controls, and a documented release handoff process.

Day 4 – Client Environment Implementation and Handover

1. Review of Training Outcomes and Open Questions
2. Assessment of the Client's Existing Environment
3. Review of Repositories, Access, Agents, and Deployment Targets
4. Mapping the Existing Development-to-Production Workflow
5. Configuration and Adaptation of the Pipeline Template
6. Pipeline Execution and Artifact Validation
7. Deployment Rehearsal in an Approved Development/Sandbox Environment
8. Release Approval, Verification, and Network-Team Handoff
9. Troubleshooting, Recovery Planning, and Outstanding Dependencies
10. Final Knowledge Transfer, Ownership, and Next Steps

Deliverable: A client-specific deployment workflow demonstration, implementation checklist, artifact handoff checklist, and action register with assigned owners.

Expected Overall Outcomes

- Develop and maintain basic Python automation scripts.
- Retrieve and process data from existing SQL-backed APIs.
- Apply basic error handling and automated testing.
- Manage code changes through Azure Repos and pull requests.
- Understand automated validation, artifact publishing, and release approvals.
- Adapt the demonstrated workflow to the client's environment.
- Establish clear deployment responsibilities, verification steps, and recovery procedures.

Delivery Approach

- **Days 1–3:** Instructor-led training, demonstrations, and hands-on exercises.
- **Day 4:** Client-environment assessment, guided implementation, deployment rehearsal, and knowledge transfer.
- **Implementation prerequisite:** Required access, permissions, pipeline agents, deployment utilities, and a nonproduction target must be available and validated in advance.

Note: Production deployment is subject to client approvals, access readiness, and the agreed implementation scope.